Rapid Intensification Prediction of North Indian Ocean Cyclones Using Machine Learning

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***Abstract*—**Rapid intensification (RI) of tropical cyclones, here defined as a gain of at least 30 kt in maximum sustained wind within 24 hours, is one of the hardest events to forecast and one of the most damaging. This work studies the India Meteorological Department (IMD) RSMC New Delhi best-track record for the North Indian Ocean, covering 425 systems and 7,583 three-hourly fixes between 1982 and 2026. The workflow covers data cleaning, exploratory analysis of eleven descriptive figures, construction of a 24-hour-ahead RI label, and five classifiers: Logistic Regression, Decision Tree, Random Forest, Gradient Boosting and Support Vector Machine. Because the label is built from future fixes, the train–test split is made by storm rather than by row. On a held-out set of 1,134 fixes containing 89 RI cases (7.85%), Gradient Boosting reached the highest accuracy (82.1%) and Random Forest and Gradient Boosting reached the highest F1-score for the RI class (0.335 and 0.334). The Decision Tree detected the most RI cases (recall 0.843) but also raised the most false alarms. Every model had precision below 0.24, so each one raises many false alarms. The results show that track and intensity history alone carry some signal for RI but are not sufficient for reliable operational use.

***Keywords*—**tropical cyclone, rapid intensification, IMD best track, class imbalance, machine learning, North Indian Ocean.

1. Introduction

The Bay of Bengal and the Arabian Sea produce only a small share of the world's tropical cyclones, but the coastal population exposed to them is very large. Forecasting the track of a cyclone has improved steadily, whereas forecasting sudden changes in intensity remains difficult. Rapid intensification, where a storm strengthens by a large amount in a short time, is the main example. A system that is forecast to be a depression can become a very severe cyclone within a day or two, leaving little time for warnings and evacuation.

The conventional threshold used in the literature is an increase of 30 kt in maximum sustained wind over 24 hours [3]. Statistical RI indices have been developed for the Atlantic and eastern Pacific basins from large-scale environmental predictors [4]. In this project the question is posed for the North Indian Ocean as a supervised classification problem: given the position, intensity and recent history of a cyclone at a particular fix, will the wind increase by at least 30 kt over the next 24 hours?

The objectives are (i) to describe the IMD best-track record through exploratory analysis, (ii) to construct a leakage-aware RI label and train–test split, (iii) to compare five standard classifiers, and (iv) to report the results with metrics suited to a rare event. RI is rare, so overall accuracy is not used as the main measure. Precision, recall and F1-score for the RI class are reported together with the confusion matrices.

2. Dataset Description

The data are the IMD RSMC New Delhi best-track record of cyclonic disturbances over the North Indian Ocean. The record was parsed with the open-source `imdtrack` Python library [1], which converts the IMD workbook into tidy data tables. According to the library documentation, the record used here contains 425 storms and 7,583 fixes for 1982–2026. Each row of the observation table is one fix of one storm, taken at roughly three-hourly intervals, and gives the time, latitude, longitude, Dvorak current-intensity number (`ci_no`), central pressure, pressure drop, maximum sustained wind in knots, the IMD intensity grade and the basin.

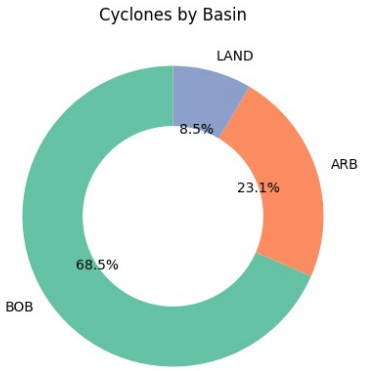


Figure 1: Share of cyclones by basin.

By basin, 291 storms belong to the Bay of Bengal (BOB, 68.5%), 98 to the Arabian Sea (ARB, 23.1%) and 36 to a LAND category (8.5%), which holds systems recorded over land (Fig. 1). The IMD grades, from weak to strong, are D (depression), DD (deep depression), CS, SCS, VSCS, ESCS and SuCS. The record for 2026 is incomplete, since the season was still in progress when the data were extracted.

3. Data Preprocessing and Exploratory Analysis

3.1 Data Cleaning

The raw IMD sheet contains some rows whose day and month are swapped. The library flags such rows, and they were corrected with its cleaning routine before analysis. Columns were then converted to the correct numeric or date types, duplicate (storm, time) rows were removed and the fixes of every storm were sorted in time order. This ordering is essential because the label is built from the wind at a later time of the same storm. Year, month and decade columns and the number of days since the start of each storm were derived.

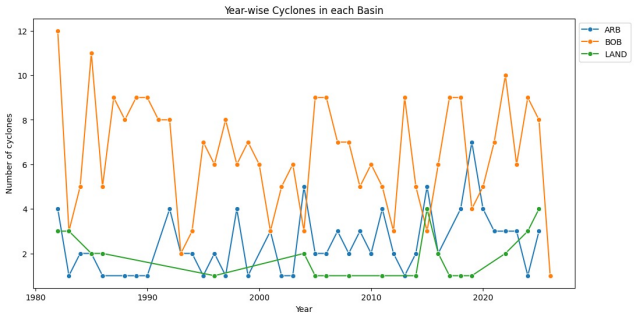


Figure 2: Year-wise number of cyclones in each basin.

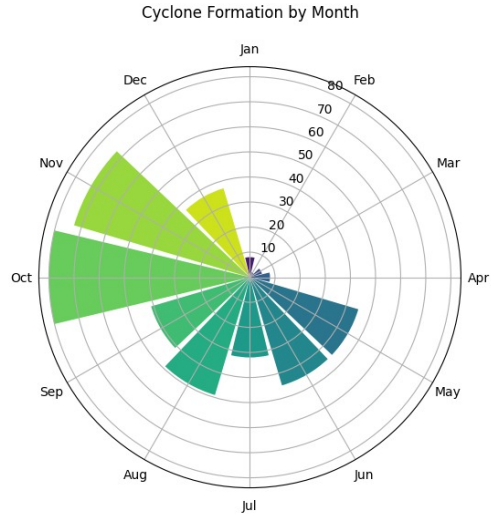


Figure 3: Cyclone formation by month.

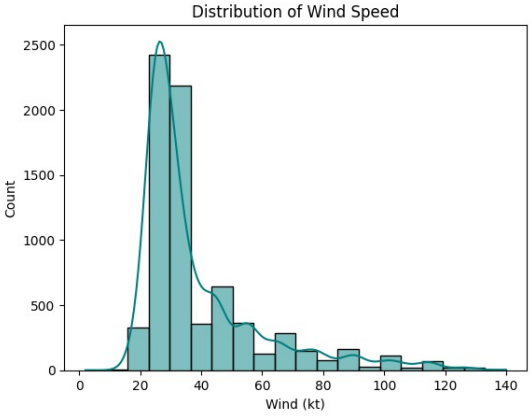


Figure 4: Distribution of wind speed over all fixes.

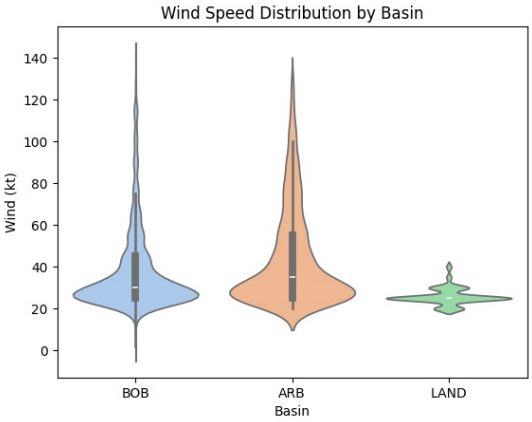


Figure 5: Wind-speed distribution by basin.

Central pressure is missing for many of the older fixes. It was not set to zero, because 0 hPa is physically impossible. Missing model inputs were instead filled with the training-set median at the modelling stage.

3.2 Cyclone Counts and Seasonality

Fig. 2 shows the yearly number of systems by basin. The Bay of Bengal has more systems than the Arabian Sea in most years. The BOB count is highest in 1982 (12 systems) and is mostly between 2 and 11 afterwards. The Arabian Sea count is mostly between 1 and 4 and peaks at 7 in 2019. The final point for 2026 is low because that year is incomplete.

The monthly distribution (Fig. 3) has two seasons. The largest peak is in October, with about 80 systems, followed by November. A smaller pre-monsoon peak appears in May, counts are moderate from June to September, and there are very few systems from January to April.

3.3 Intensity Characteristics

The wind-speed distribution (Fig. 4) is strongly right-skewed. Most fixes lie around 25–35 kt, which corresponds to the depression stage, and the frequency falls quickly for stronger winds, with a thin tail up to about 140 kt. Strong winds are therefore rare, and so are large intensity gains. The violin plot (Fig. 5) shows the same pattern for both open-sea basins. The BOB and ARB distributions are both concentrated near 30 kt but extend above 100 kt, and the ARB median is slightly higher. LAND fixes are confined to low winds below about 40 kt, as expected after landfall.

Fig. 6 stacks the number of fixes in each IMD grade by decade. The count is largest in the 2010s, while the 1980s are lower because the record starts in 1982 and the 2020s are an incomplete decade. Weak grades (D and DD) dominate in every decade and the high grades VSCS, ESCS and SuCS form a thin top layer. The box plot of the maximum wind of each storm (Fig. 7) has a median close to 30 kt in every decade. The 1990s have the highest upper quartile and the strongest storm in the record, at 140 kt. This storm, labelled 1999-007, is the strongest in the data set; AMPHAN (2020), KYAAR (2019), 1990-001 (1990) and GONU (2007) follow it. No clear monotonic trend in storm strength by decade can be read from the box plots.

3.4 Spatial Patterns

The genesis density map (Fig. 8) shows two clusters. The strongest is in the northern Bay of Bengal near 88–90°E and 18–22°N, with a broader band extending south to about 10°N. A weaker cluster lies in the eastern Arabian Sea between about 65°E and 72°E. The map of all fixes (Fig. 9) colours each fix by wind. The strongest winds lie along curved tracks over open water in both basins, while the weak fixes cluster near the head of the Bay of Bengal and near the coast, where storms weaken after landfall.

3.5 Life Cycle of Strong Storms

Fig. 10 plots the wind history of the five strongest storms against days since formation. All five rise from weak intensity (25–45 kt) to above 90 kt within about two to three days, and several gain tens of knots between consecutive days. This steep rise is the behaviour that the RI label is designed to capture. The peak is followed by a decline, after landfall or over cooler water, which is longest for KYAAR (2019).

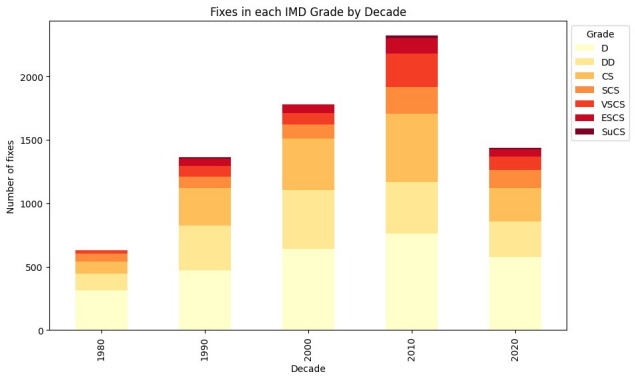


Figure 6: Number of fixes in each IMD grade by decade.

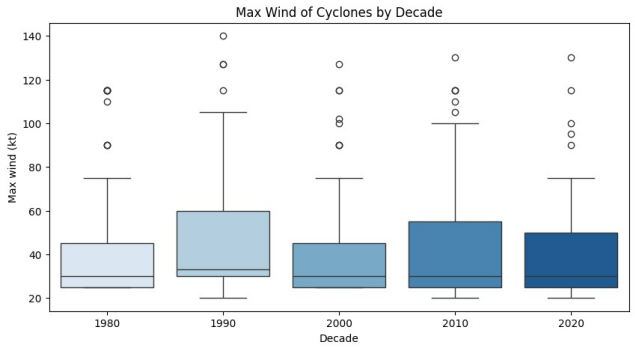


Figure 7: Maximum wind of cyclones by decade.

4. Machine-Learning Methodology

4.1 Rapid-Intensification Label

Fixes are not equally spaced in time (for example 00, 03, 06, 12 and 18 UTC), so the next row cannot be used as the 24-hour future. For each fix the wind of the same storm exactly 24 h later was found by shifting the time stamps and merging. The label is

$$RI = \begin{cases} 1, & w(t + 24\,\mathrm{h}) - w(t) \geq 30\ \mathrm{kt} \\ 0, & \text{otherwise,} \end{cases} \tag{1}$$

where w is the maximum sustained wind. Fixes for which no fix exists exactly 24 h later are dropped. In the same way, the wind 24 h earlier was found and the difference $w(t) - w(t - 24\,\mathrm{h})$, called `past_change`, was used as an input. It uses past information only, so it does not leak the label. Missing values of this feature were set to zero.

4.2 Features and Pre-processing

Ten inputs were used: current wind, central pressure, pressure drop, Dvorak CI number, latitude, longitude, month, basin code, grade code and `past_change`. IMD grades were encoded as integers from 0 (D) to 6 (SuCS). The basin was coded as BOB = 0, ARB = 1 and any other value, which includes LAND, as 2. Missing values were filled with the median and the features were standardised. Both steps were fitted on the training data only and then applied to the test data.

4.3 Storm-Wise Train–Test Split

Consecutive fixes of one storm are highly similar, and the label of a fix depends on later fixes of the same storm. A random split over rows would put neighbouring fixes of one storm in both the training and the test set and would give optimistic scores. A grouped split with the storm identifier

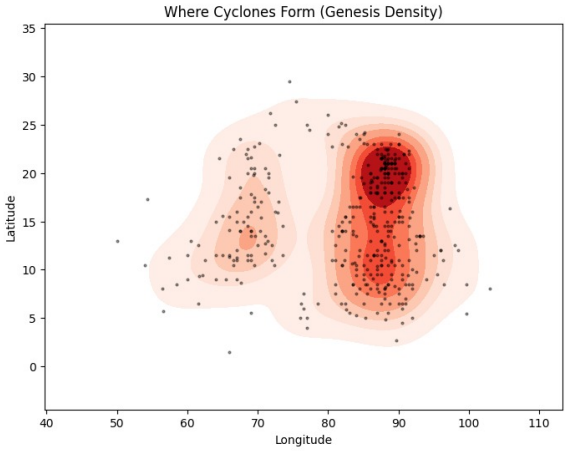


Figure 8: Genesis density of North Indian Ocean cyclones.

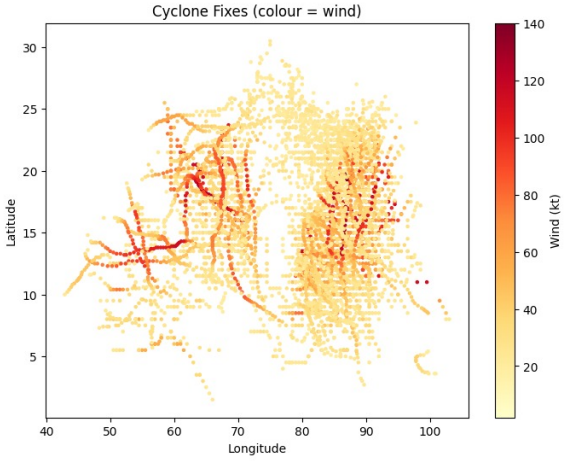


Figure 9: All cyclone fixes coloured by wind speed (kt).

as the group was therefore used (`GroupShuffleSplit`, 25% of storms for testing, random state 42). The notebook checks that no storm appears in both sets. The resulting test set has 1,134 fixes of which 89 (7.85%) are RI.

4.4 Models

Five classifiers from scikit-learn [5] were trained. Since RI is rare, class weights were used to give the RI class more importance (inverse-frequency weights).

- **Logistic Regression** [6]: balanced class weights, up to 1000 iterations.
- **Decision Tree** [7]: maximum depth 5, at least 10 samples per leaf, balanced class weights.
- **Random Forest** [8]: 200 trees, maximum depth 8, at least 5 samples per leaf, balanced class weights.
- **Gradient Boosting** [9]: 150 trees, learning rate 0.1, depth 3, trained with balanced sample weights because the estimator has no class-weight option.
- **Support Vector Machine** [10]: RBF kernel, $C = 1$, balanced class weights.

4.5 Evaluation Measures

With about 92% of the test fixes being non-RI, a model that always predicts "no RI" would score about 92% accuracy and detect nothing. Accuracy is therefore reported only for completeness. The main measures are precision $= TP/(TP + FP)$, recall $= TP/(TP + FN)$ and their

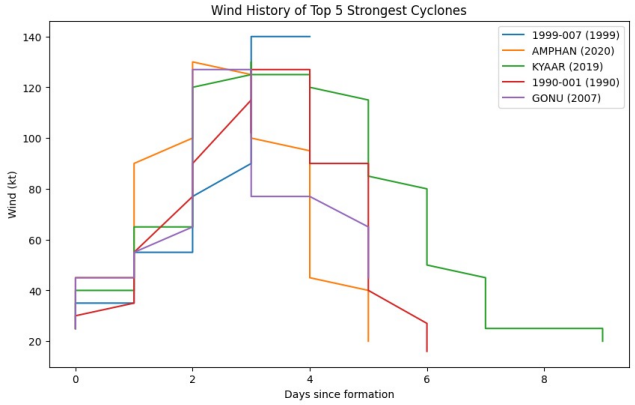


Figure 10: Wind history of the five strongest cyclones.

Table 1: Test-set results for the RI class (1,134 fixes, 89 RI).

Model	Acc.	Prec.	Recall	F1	Spec.
Logistic Regression	0.647	0.144	0.708	0.240	0.642
Decision Tree	0.626	0.155	**0.843**	0.261	0.608
Random Forest	0.787	0.222	0.685	**0.335**	0.795
Gradient Boosting	**0.821**	**0.236**	0.573	0.334	**0.842**
SVM (RBF)	0.694	0.164	0.708	0.266	0.693

harmonic mean, the F1-score, all for the RI class. Specificity $= TN/(TN + FP)$ is added to show the false-alarm side. All numbers below were computed from the confusion matrices in Fig. 11.

5. Results and Discussion

Table 1 lists the results on the storm-wise test set, and Fig. 11 shows the five confusion matrices.

The models fall into two groups. Logistic Regression, the Decision Tree and the SVM catch 63 to 75 of the 89 RI fixes, but they also raise 321 to 410 false alarms, so their precision is only 0.14 to 0.16. Random Forest and Gradient Boosting are more conservative. They raise 214 and 165 false alarms and detect 61 and 51 RI fixes. As a result they reach the best F1-scores, 0.335 and 0.334, which are almost identical, and the best accuracy and specificity. The Decision Tree has the highest recall (75 of 89, 0.843), but at the cost of the most false alarms (410).

No model is clearly best on all measures. The choice depends on the cost of a missed event against the cost of a false alarm. For a warning system, in which missing RI is much worse than an extra alert, the Decision Tree or the SVM and Logistic Regression would be preferred. If false alarms must be limited, Random Forest or Gradient Boosting are better. Since the test set has only 89 positive fixes, which come from a limited number of storms, small differences such as that between 0.335 and 0.334 should not be interpreted as a ranking.

The base rate of RI in the test set is 7.85%. The precision of the best models (0.22 and 0.24) is about three times that value, so the models do contain real signal, but about three out of four alerts are still false. The low precision is also partly a consequence of the balanced class weights, which move the decision threshold towards the RI class. No threshold tuning or probability calibration was done, and ROC-AUC and cross-validated scores are not reported here.

The descriptive analysis supports the choice of inputs. RI happens at the intensifying stage of a storm, which the wind, pressure, grade and `past_change` variables describe. The monthly and spatial figures show that both the season and the region are concentrated, so month, latitude and longitude carry information as well.

6. Limitations

Several limitations affect the interpretation of the results.

- **No environmental predictors.** RI is driven mainly by sea-surface temperature, ocean heat content, vertical wind shear and humidity. None of them is in the best-track record, so the models rely on track and intensity information only.
- **Small number of positive cases.** The test set holds only 89 RI fixes, and fixes of the same storm are strongly correlated, so the effective sample size is smaller than the row count.
- **Single split.** The results come from one storm-wise split. Repeating the evaluation with grouped cross-validation would show how much the scores depend on which storms fall into the test set.
- **Missing pressure and imputation.** Central pressure is missing for many older fixes and was imputed with the median, which can blur the relation between pressure and wind.
- **LAND category.** Fixes over land are included. They are weak by definition, and they share one code with any unmapped basin value.
- **Best-track data quality.** The best track is a post-analysis product whose intensity estimates in older years are based on satellite estimates and are less accurate. The fixes are also not equally spaced, so only fixes with an exact 24-hour future were used.
- **Incomplete 2026.** The last year of the record is incomplete.

7. Conclusion

This project built a pipeline for predicting 24-hour rapid intensification of North Indian Ocean cyclones from the IMD best-track record of 425 systems and 7,583 fixes. Exploratory analysis showed a bimodal season with a peak in October, genesis concentrated in the northern Bay of Bengal, and a strongly skewed wind distribution in which strong storms are rare. The RI label was built from the exact 24-hour future, and the evaluation used a storm-wise split to avoid leakage between neighbouring fixes of one storm.

Among the five classifiers, Random Forest and Gradient Boosting gave the best balance (F1 about 0.33), the Decision Tree detected the largest share of RI cases (recall 0.84), and all models had low precision. The results show that intensity and position history carry a measurable but limited signal for RI.

Future work should add sea-surface temperature, ocean heat content and wind shear from reanalysis data, tune the decision threshold with precision–recall analysis, use grouped cross-validation with several random splits, and test a chronological hold-out that trains on early years and tests on the latest ones.

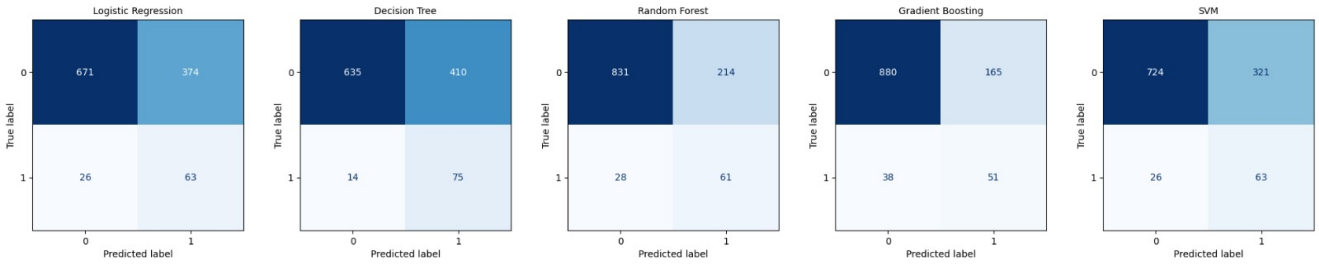


Figure 11: Confusion matrices of the five models on the storm-wise test set (0 = no RI, 1 = RI).

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